

Bradley Pfeil

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Personal Site ✦ LinkedIn ✦ GitHub

Education

The University of Adelaide

2022 – 2024

Bachelor of Computer Science (Advanced)

Work Experience

Data Scientist – Streamline (Lunio)

2025 – Current

Worked with a team to build a dynamic pricing platform for enterprise clients that surfaces revenue opportunities through demand modelling and optimisation. Personally modelled transition probabilities with a two-stage approach (hurdle classifier plus a gradient-boosted regressor with uncertainty estimates) and jointly developed a Markov Decision Process solver that recommends optimal prices, forecasts revenue. Estimated how customers respond to price changes using causal inference, and added a counterfactual data pipeline so the models generalise beyond the price points seen in the data. Currently developing a deep Q-learning agent (PyTorch) that treats commodity-storage pricing as a sequential control problem the agent learns to optimise. Shipped models to production on AWS with CI/CD pipelines. Refactored codebase to optimise bottlenecks, halving inference time. Able to cut monthly AWS spend by 20%+ through a dedicated cost-monitoring framework. Worked closely with clients to explain results, shape reward functions around their goals, and keep pricing recommendations grounded in the business. Started a paper-reading group to keep the team current on ML research.

Intern / Graduate Data Scientist – Aurizn

2023 – 2025

Built LLM-based recommender systems consuming Shapley values to explain their decisions in terms non-technical stakeholders could act on. Worked with the team to build client-requested features and model improvements for a dynamic pricing system, including data ingestion and automated retraining that kept accuracy steady as the data shifted and cut the manual upkeep the team had to do.

Projects

Adaptive Database Optimiser

2025

Built a self-tuning system that watches SQL query workloads and automatically retunes how Parquet data is partitioned and sorted to cut query latency and I/O. It logs each query's access patterns, proposes new layouts from workload analysis, rewrites the data, and routes traffic across layout versions using multi-armed and contextual bandits (UCB1, Thompson sampling), learning which layouts win as access patterns shift. Built in Python with DuckDB and PyArrow, with statistically gated rewards so only confident improvements are kept.

QueryGPT: NL-to-SQL Pipeline

2025

Built a multi-agent pipeline that turns plain-English questions into SQL through four staged agents: intent classification, table selection, column pruning, and SQL generation. Progressively narrowing the schema context at each stage cuts token usage and reduces failure cases, and 'pydantic-ai' structured outputs with automatic retries keep each step reliable. Used 'llama-index' and a Neo4j hybrid vector store to retrieve similar query examples (RAG), and validated generated SQL end-to-end so errors surface immediately.

Languages & tools

Proficient: Python, SQL, GBT, Polars, PyTorch, AWS, Docker, Git, Linux

Experienced: PySpark, Rust

Additional Experience

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- Scrum Master Certificate 2025